

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course: B.TECH****Branch : Electrical Engineering****Semester : VI****Subject Code & Name: BTEEC605A_Y19 & Elective-VII A. Switch Gear and Protection****Max Marks: 60****Date: 10/07/25****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Discrimination between main and back up protection is provided by the use of relays which are a. fact b. sensitive c. slow d. none of the above		1
2	Directional relays are based on flow of a. power b. current c. voltage wave d. all of the above		1
3	Admittance relay is _____ relay a. impedance b. directional c. non-directional d. none of the above		1
4	The voltage appearing across the contacts after opening of the circuit breaker is called _____ voltage a. recovery b. surge c. operating d. arc		1
5	Inductance cup relay is operated due to changes in a. current b. voltage c. impedance d. all of the above		1
6	Arc in a circuit breaker is interrupted at a. zero current b. maximum current c. maximum voltage d. minimum voltage		1
7	A transmission line is protected by a. inrush protection b. distance protection c. time graded and graded d. both (b) and (c)		1

			over current protection			
8	A differential relay measures the vector difference between					1
	a. two currents	b. two voltages	c. two or more similar electrical quantities	d. none of the above		
9	For which of the following protection from negative sequence currents is provided					1
	a. generators	b. motors	c. transmission line	d. transformers		
10	For the protection of stator winding of an alternate against internal fault involving ground the relay used is a					1
	a. Biased differential relay	b. Directional over-current relay	c. Plain impedance relay	d. Buchholz relay		
11	The under-voltage relay can be used for					1
	a. Generators	b. Bus bars	c. Transformers	d. All of the above		
12	The arc resistance depends on which among the following factors					1
	a. Cross section of the arc.	b. Length of the arc.	c. Degree of ionization	d. All of the above.		
Q. 2	Solve the following.					12
A)	Discuss primary and backup protection system.					6
B)	Explain SF6 circuit breaker					6
Q.3	Solve the following.					12
A)	Explain principle of working of induction cup type relay					6
B)	Explain the working principle of differential relays.					6
Q. 4	Solve Any Two of the following.					12
A)	Explain minimum oil circuit breaker.					6
B)	Explain Buchholz relay with neat diagram.					6
C)	Explain the working principle of distance relays.					6

Q.5	Solve Any Two of the following.		12
A)	Explain construction working lightning arrester and its application.		6
B)	Explain the following: (i) Flashover voltage (ii) Basic Impulse Insulation		6
C)	Explain the principle and operation of Vacuum Breaker with diagram		6
Q. 6	Solve Any Two of the following.		12
A)	Explain the insulation coordination.		6
B)	Explain the circulating current protection of busbar using over current relay.		6
C)	Discuss the percentage differential protection scheme of a transformer		6
	*** End ***		